MILITARY SPECIFICATION

HOSE, SYNTHETIC RUBBER, SYNTHETIC FIBER REINFORCED, AND END FITTINGS, REUSABLE, FOR FLEXIBLE HOSE ASSEMBLIES (LOW PRESSURE SIZES -80 through -192) AND ADAPTERS

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers synthetic fiber reinforced hoses and fittings in sizes -80 through -192 for low pressure applications.

1.2 Classification. End fittings for hose covered by this specification shall be in sizes -80 through -192, as applicable, and shall be of the following types, as specified (see 6.2).

Type AI - In accordance with figure 1, with 400 lb/in^2 flange (see figure 1 of MIL-H-24136).
Type AII - In accordance with figure 1, with 150/250 lb/in^2 flange (see figure 1 of MIL-H-24136).
Type AIII - In accordance with figure 1, with 250 lb/in^2 flange (see figure 1 of MIL-H-24136).
Type AIV - In accordance with figure 1, with 150 lb/in^2 flange (see figure 1 of MIL-H-24136).
Type AV - In accordance with figure 1, with 150 lb/in^2 B16.5 flange (see figure 1 of MIL-H-24136).
Type BI - In accordance with figure 2, with 400 lb/in^2 flange (see figure 1 of MIL-H-24136).
Type BII - In accordance with figure 2, with 150/250 lb/in^2 flange (see figure 1 of MIL-H-24136).
Type BIII - In accordance with figure 2, with 250 lb/in^2 flange (see figure 1 of MIL-H-24136).
Type BIV - In accordance with figure 2, with 150 lb/in^2 flange (see figure 1 of MIL-H-24136).
Type BV - In accordance with figure 2, with 150 lb/in^2 B16.5 flange (see figure 1 of MIL-H-24136).
Type C - In accordance with figures 3 and 7 or 8 and 9.
Type D - In accordance with figures 4 and 7 or 8 and 9.
Type E - In accordance with figure 5.
Type F - In accordance with figure 6.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4730

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications. Unless otherwise specified, the following specifications of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-B-654 - Brazing Alloys, Silver.
QQ-C-390 - Copper Alloy Castings, (Including Cast Bar).
QQ-C-591 - Copper-Silicon, Copper-Zinc-Silicon, and Copper-Nickel-Silicon Alloys: Rod, Wire, Shapes, Forgings, and Flat Products (Flat Wire, Strip, Sheet, Bar, and Plate).
QQ-N-281 - Nickel-Copper Alloy Bar, Rod, Plate, Sheet, Strip, Wire, Forgings, and Structural and Special Shaped Sections.
QQ-N-286 - Nickel-Copper-Aluminum Alloy, Wrought.
QQ-N-288 - Nickel-Copper Alloy and Nickel-Copper-Silicon Alloy Castings.

MILITARY

MIL-H-24136 - Hose, Synthetic Rubber, Synthetic Fiber Reinforced and End Fittings, Reusable, for Flexible Hose Assemblies, General Specification for.
MIL-B-24480 - Bronze, Nickel-Aluminum Castings, for Seawater Service.

(Copies of specifications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Hose and end fittings furnished under this specification shall conform to the requirements of MIL-H-24136 and as specified herein.

3.2 Dimensions and pressures. Hose dimensions and pressures shall be in accordance with table I.

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TABLE I. Dash number, performance characteristics and dimensions.

Dash no. (hose size)	Pressure (lb/in^2)		Hose dimensions (inches)				Hose reinforcement		Minimum bend radius (inches)	Impulse peak pressure (lb/in^2)
	Working pressure	Burst pressure	I.D.		O.D.		Construction	No. of layers		
			Min	Max	Min	Max				
-80	300	1,200	4.985	5.061	5.690	5.878	Spiral	6	36	375
-96	300	1,200	4.985	6.061	6.750	6.938	Spiral	6	42	375
-128	300	1,200	7.985	8.061	8.781	8.969	Spiral	8	54	375
-160	200	800	9.985	10.061	10.781	10.969	Spiral	8	66	250
-192	150	600	11.985	12.061	12.925	13.249	Spiral	10	78	188

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance provisions shall be in accordance with MIL-H-24136 and as specified herein.

4.2 Sampling for qualification testing.

4.2.1 Hose. Qualification testing of hose shall consist of hose samples as specified in table II. Samples shall be not less than 15 inches in length and shall be fitted with qualified end fittings from at least two previously qualified suppliers (see 4.2.4).

TABLE II. Numbers of hose samples for qualification testing.

Hose size (dash no.)	Number of hose samples to be provided
-80	3
-128	4
1/	3

1/ This sample shall represent the largest size hose or fitting proposed for qualification.

4.2.2 End fittings. Qualification testing of end fittings only, shall consist of 20 end fittings of the same type as specified in table III. End fitting samples shall be installed on 10 qualified hose samples, in the sizes listed in table II, from at least two previously qualified hoses (see 4.2.4).

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TABLE III. Numbers of end fitting samples for qualification testing.

End fitting size (hose dash no.)	Number of end fitting samples to be provided
-80	6
-128	8
1/	6

1/ This sample shall represent the largest size hose or fitting proposed for qualification of larger than -128.

4.2.3 Hose and end fittings combined. In the event that a single contractor of both hose and fittings desires to qualify both hose and fittings, or a hose contractor and a fitting contractor desire to qualify hose and fittings in a combined qualification test, the samples shall be provided as specified in table IV.

4.2.4 Sample requirements when there are less than two previously qualified suppliers. Where there is only one qualified source for qualified samples, all the previously qualified samples required shall come from that source. Where there are no previously qualified sources, all samples of hose and fittings shall be submitted by the prospective qualifier.

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TABLE IV. Number of samples for combined qualification testing.

One approved source					
Hose size (dash no.)	No. of hose samples to be provided		End fitting (hose dash no.)	No. of end fittings to be provided	
	Qualifier 2/	Approved 1/ source		Qualifier 2/	Approved 1/ source
-80	2	1	-80	4	2
-128	2	2	-128	4	4
3/4/	2	1	3/	4	2

Two or more approved sources						
(Hose size dash no.)	Number of hose samples to be provided by			End fitting size (dash no.)	Number of end fittings to be provided by		
	Qualifier 2/	Approved 1/ source #1	Approved 1/ source #2		Qualifier 2/	Approved 1/ source #1	Approved source #2
-80	2	1	0	-80	4	2	0
-128	2	1	1	-128	4	2	2
3/4/	2	1	0	3/4/	4	2	0

Hose samples shall be not less than 15 inches in length and shall be coupled on one end with the fitting being submitted for qualification and on the other end with previously qualified end fittings.

1/ Approved source - Previously qualified hose and fittings, or hose or fitting.
2/ Qualifier - Hose and fittings, or hose or fitting to be qualified.
3/ Satisfactory completion of qualification testing of the samples selected in accordance with tables II, III and IV will qualify the manufacturer for all sizes between the largest size tested and -80.
4/ If the largest size proposed is -128, then the number of hose samples shall equal 7 and fittings 14. If a size larger than -128 is proposed, the total number of hose samples shall be 10 and fittings 20.

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4.2.5 Sequence of tests on sample hose assemblies shall be as specified in table V.

TABLE V. Sequence of qualification tests on sample assemblies.

Test sequence	Sample numbers									
	Size (dash no.) -80			Size (dash no.) -128				Largest size over -128 2/		
	1	2	3	4	5	6	7	8	9	10
Proof pressure (see 4.6.1) 1/	X	X	X	X	X	X	X	X	X	X
Stability test (see 4.6.2) 1/	X	X	X	X	X	X	X	X	X	X
Aging test (see 4.6.3) 1/		X	X		X	X	X	X	X	X
Hydraulic fluid circulation (see 4.6.4) 1/		X			X				X	
Impulse test (see 4.6.5) 1/			X				X			X
Burst test (see 4.6.6) 1/	X			X				X		
Vacuum test (see 4.6.7) 1/			X				X			
Cold temperature flexing (see 4.6.8) 1/						X				X
Adhesion test (see 4.6.9) 1/			X				X			

1/ See MIL-H-24136.
2/ Largest size sample submitted for qualification.

5. PACKAGING

5.1 Preparation for delivery requirements shall be in accordance with MIL-H-24136.

6. NOTES

6.1 Intended use. Hose and fittings to this specification are intended for use in shipboard or shore to ship piping systems where pressures do not exceed the rated working pressure, and temperatures do not exceed minus 40°F or 200°F.

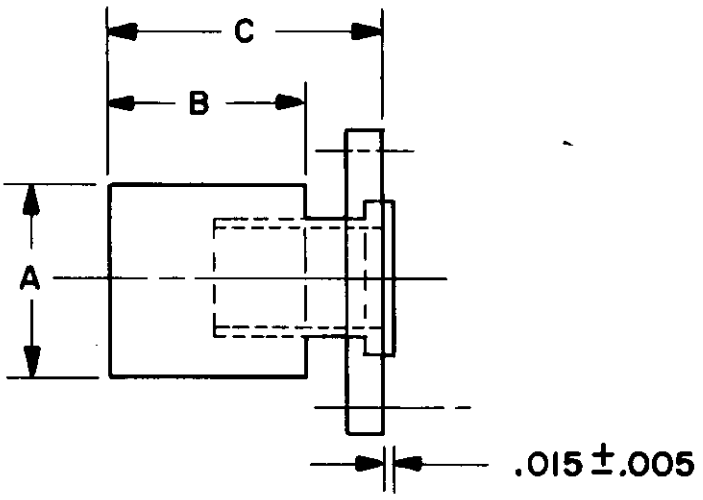
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6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Type of end fittings for hoses (see 1.2).

Preparing activity:
Navy - SH
(Project 4730-N874-1)

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Hose size (dash no.)	A 1/	B 1/	C 1/
-80	11.50	5.00	7.00
-96	12.00	5.00	7.00
-128	14.00	5.00	7.00
-160	16.50	5.00	7.50
-192	20.00	6.50	9.00

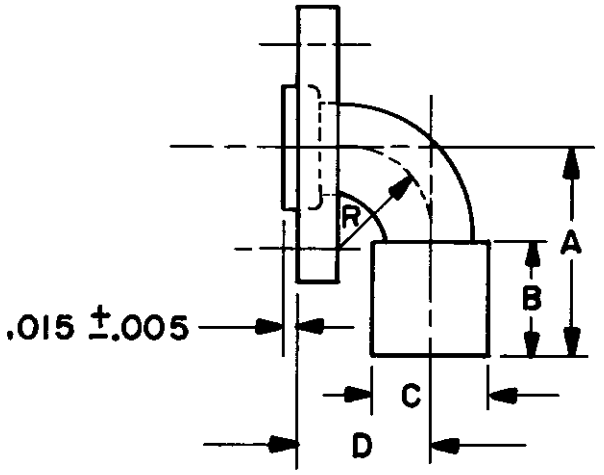
1/ Maximum dimensions.

SH 12285

NOTES:
1. Dimensions are in inches.
2. Materials: All parts in direct contact with sea water shall be in accordance with MIL-B-24480.
Bolts and nuts - QQ-N-281, QQ-N-286.
All other parts - QQ-N-281, QQ-N-286, QQ-N-288 or MIL-B-24480.

FIGURE 1. Types AI, AII, AIII, AIV or AV, end fitting hose to flange, straight.

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Hose size (dash no.)	A 1/	B 1/	C 1/	D 1/	R (nom)
-80	8.75	5.00	11.50	7.00	3.50
-96	9.25	5.00	12.00	7.50	4.00
-128	10.50	5.00	14.00	9.00	5.00
-160	12.00	5.00	16.50	10.00	6.38
-192	15.00	6.50	20.00	11.00	7.38

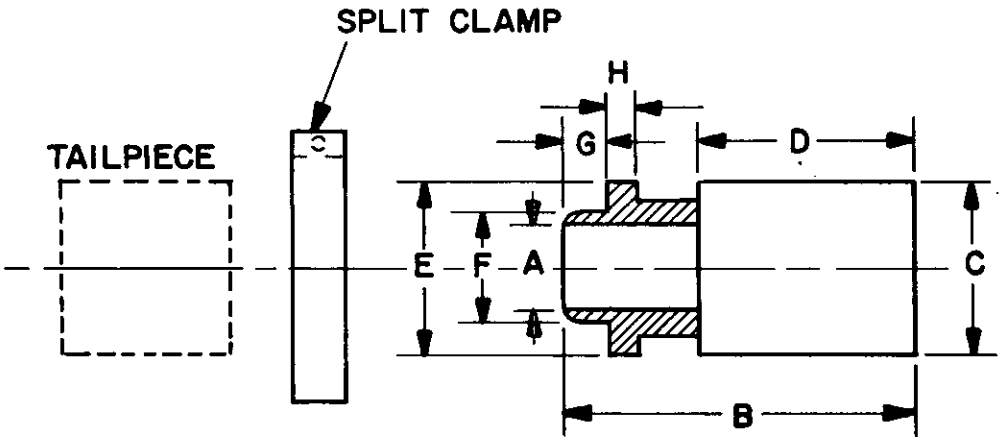
1/ Maximum dimensions.

SH 12286

NOTES:
1. Dimensions are in inches.
2. Materials: All parts in direct contact with sea water shall be in accordance with MIL-B-24480.
Bolts and nuts - QQ-N-281, QQ-N-286.
All other parts - QQ-N-281, QQ-N-286, QQ-N-288 or MIL-B-24480.

FIGURE 2. Types BI, BII, BIII, BIV or BV, end fitting hose to flange, 90 degree elbow.

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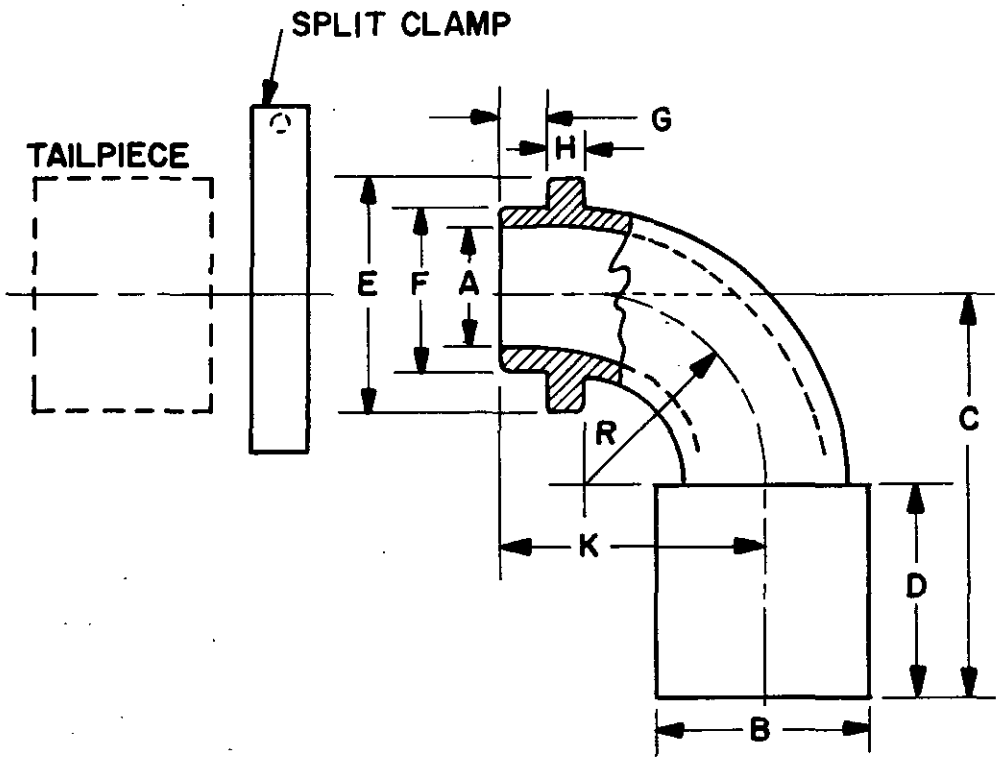
Hose size (dash no.)	A (max)	B (max)	C (max)	D (max)	E ±0.006	F +0.006 −0.000	G +0.005 −0.000	H +0.005 −0.000
-80	5.00	6.25	11.50	5.00	6.63	5.62	0.40	0.36
-96	6.00	6.75	12.00	5.00	7.57	6.75	0.45	0.43
-128	8.00	7.00	14.00	5.00	9.97	9.00	0.82	0.40
-160	10.00	7.00	16.50	5.00	12.06	11.00	0.86	0.40
-192	12.00	9.00	20.00	6.50	13.38	12.00	0.88	0.44

SH 12287

NOTES:
1. Dimensions are in inches.
2. Materials: All parts in direct contact with sea water shall be in accordance with MIL-B-24480.
Bolts and nuts - QQ-N-281, QQ-N-286.
All other parts - QQ-N-281, QQ-N-286 or QQ-N-288.

FIGURE 3. Type C, end fittings, straight split clamp with braze tailpiece and clamp assembly.

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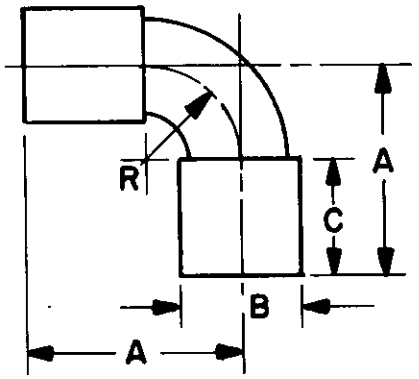
SH 12289

Hose size (dash no.)	A (max)	B (max)	C (max)	D (max)	E ±0.006	F +0.006 -0.000	G +0.005 -0.000	H +0.005 -0.000	R	K (max)
-80	5.00	11.50	8.75	5.00	6.63	5.62	0.40	0.36	3.50	7.00
-96	6.00	12.00	9.25	5.00	7.57	6.75	0.45	0.43	4.00	7.50
-128	8.00	14.00	10.50	5.00	9.97	9.00	0.82	0.40	5.00	9.00
-160	10.00	16.50	12.00	5.00	12.06	11.00	0.86	0.40	6.38	10.00
-192	12.00	20.00	15.00	6.50	13.38	12.00	0.88	0.44	7.38	11.00

NOTES:
1. Dimensions are in inches.
2. Materials: All parts in direct contact with sea water shall be in accordance with MIL-B-24480.
   Bolts and nuts - QQ-N-286.
   All other parts - QQ-N-281, QQ-N-288, MIL-B-24480 or QQ-N-286.

FIGURE 4. Type D, end fittings 90 degrees with split clamp and brazed tailpiece and split clamp.

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Hose size (dash no.)	A 1/	B 1/	C 1/	R 1/
-80	8.75	11.50	5.00	3.50
-96	9.25	12.00	5.00	4.00
-128	10.50	14.00	5.00	5.00
-160	12.00	16.50	5.00	6.38
-192	15.00	20.00	6.50	7.38

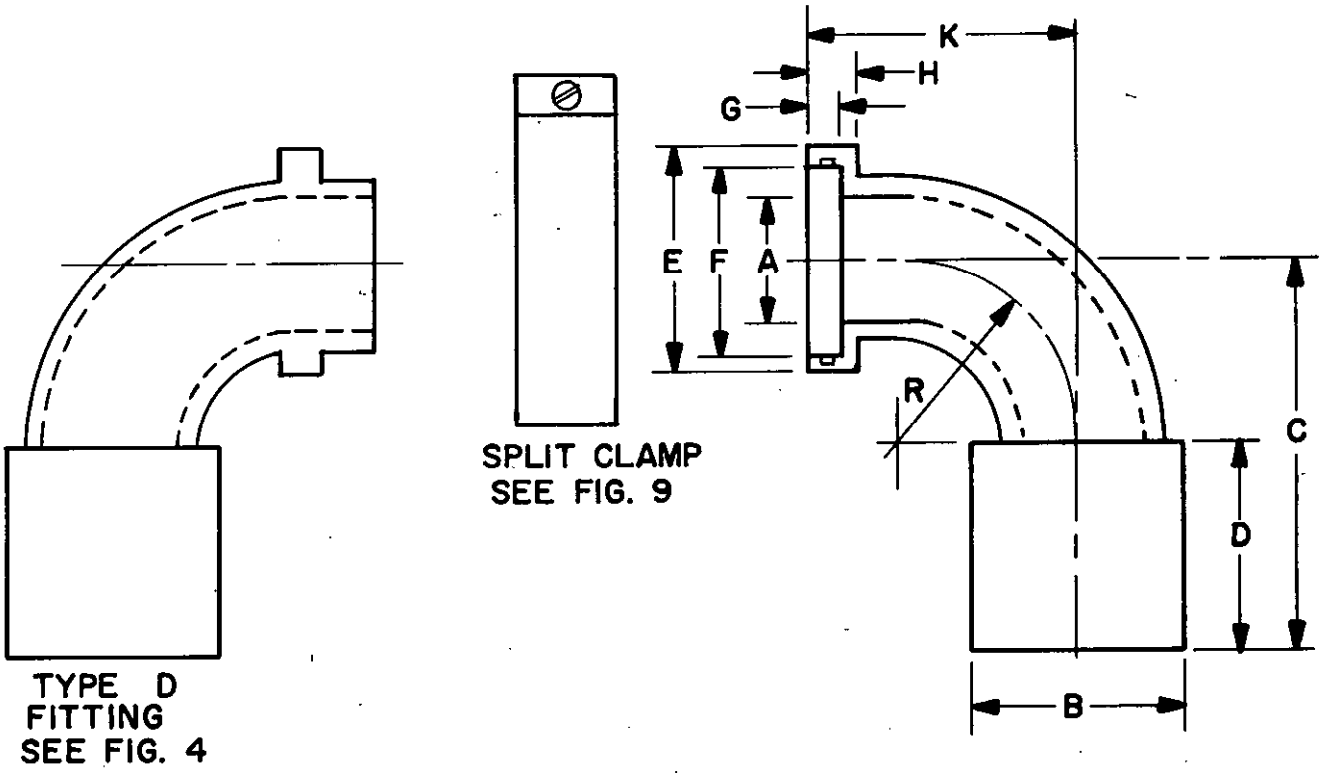
1/ Maximum dimensions.

SH 12290

NOTES:
1. Dimensions are in inches.
2. Materials: All parts in direct contact with sea watr shall be in accordance with MIL-B-24480.
   Bolts and nuts - QQ-N-281, QQ-N-286.
   All other parts - QQ-N-281, QQ-N-288, MIL-B-24480 or QQ-N-286.

FIGURE 5. Type E fitting, 90 degree elbow.

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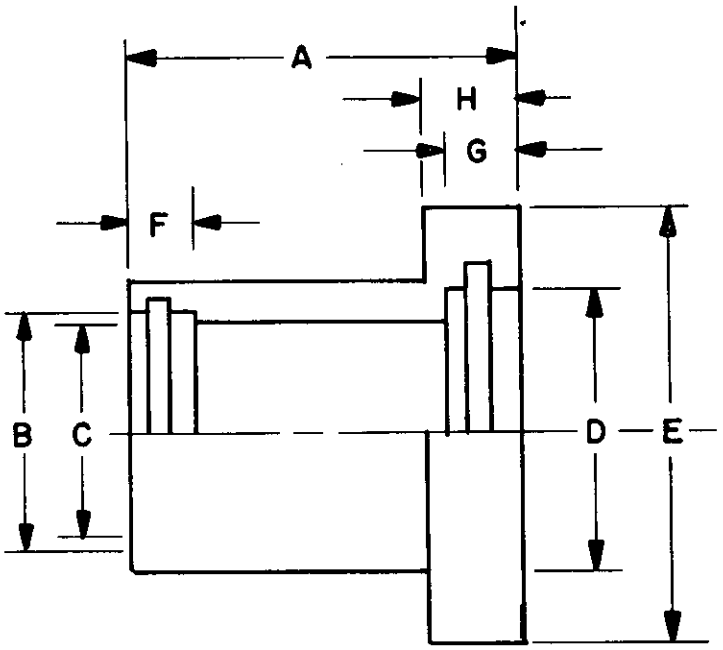
Hose size (dash no.)	A NPS (nom)	B (nom)	C (nom)	D (nom)	E ±0.006	F +0.006 −0.000	G +0.005 −0.000	H +0.005 −0.000	R (nom)	K (nom)
-80	5.00	11.50	8.75	5.00	6.63	5.64	0.42	0.36	3.50	7.00
-96	6.00	12.00	9.25	5.00	7.57	6.76	0.47	0.43	4.00	7.50
-128	8.00	14.00	10.50	5.00	9.97	9.01	0.84	0.40	5.00	9.00
-160	10.00	16.50	12.00	5.00	12.06	11.01	0.88	0.40	6.38	10.00
-192	12.00	20.00	15.00	6.50	13.38	12.01	0.90	0.44	7.35	11.00

SH 12291

NOTES:
1. Dimensions are in inches.
2. Materials: All parts in direct contact with sea water shall be in accordance with MIL-B-24480.
   Bolts and nuts - QQ-N-281, QQ-N-286.
   All other parts - QQ-N-281, QQ-N-288, MIL-B-24480 or QQ-N-286.

FIGURE 6. Type F, end fitting, for 180 degree return configuration with split clamp.

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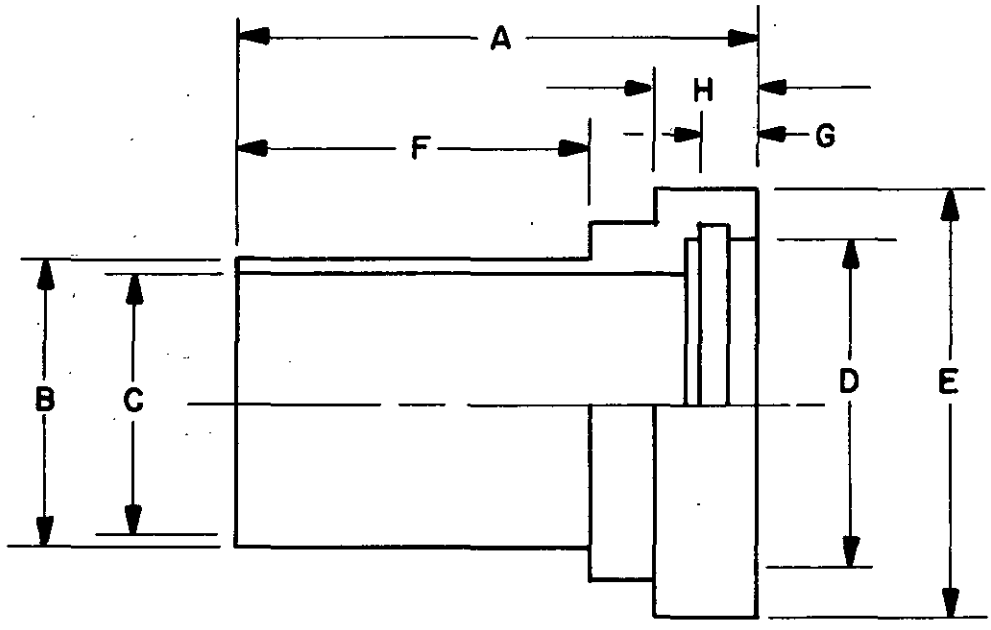
Hose size (dash no.)	A (max)	B +0.006 -0.000	C (nom)	D +0.006 -0.000	E +0.006 -0,000	F +0.006 -0.000	G +0.005 -0.000	H +0.005 -0.000
-80	1.80	6.25	5.30	5.64	6.63	1.00	0.42	0.36
-96	1.95	5.57	6.63	6.76	7.57	1.11	0.47	0.43
-128	2.50	6.63	8.40	9.01	9.97	1.31	0.84	0.40
-160	2.75	8.81	10.53	11.01	12.06	1.49	0.88	0.40
-192	3.00	10.75	12.58	12.01	13.38	3.10	0.90	0.40

SH 12292

NOTES:
1. Dimensions are in inches.
2. Material: MIL-B-24480
   O-ring - buna N
   Braze ring - QQ-B-654

FIGURE 7. Sil-braze tailpiece, female.

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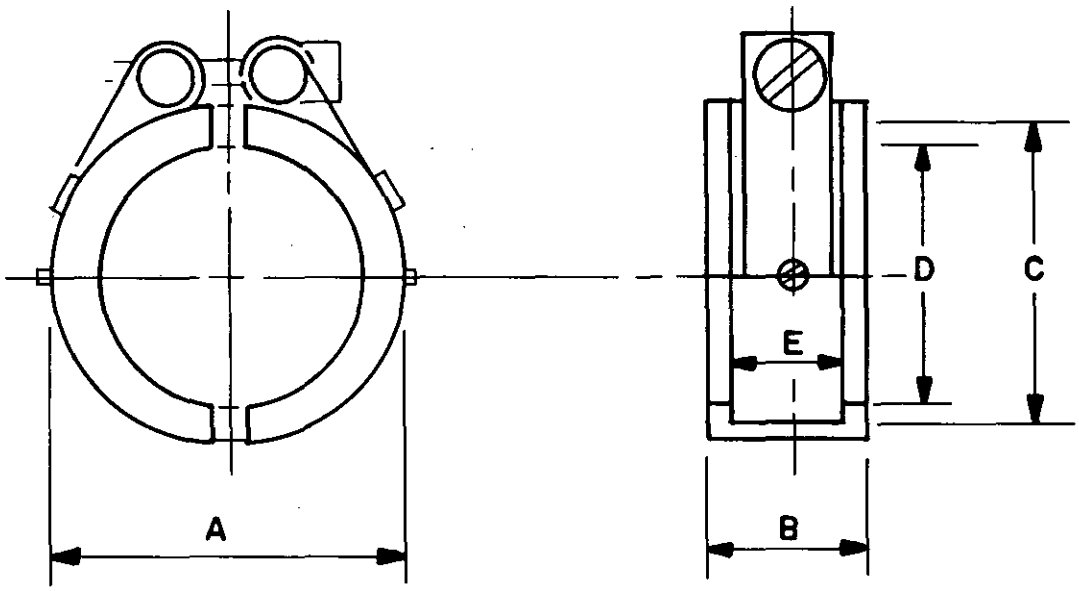
Hose size (dash no.)	A (max)	B +0.000 -0.006	C (nom)	D +0.006 -0.000	E ±0.006	F (nom)	G +0.005 -0.000	H +0.005 -0.000
-80	3.80	5.56	4.94	5.64	6.63	2.77	0.42	0.36
-96	3.87	6.62	5.96	6.76	7.57	2.67	0.47	0.48
-128	4.37	8.62	7.88	9.01	9.97	2.82	0.84	0.40
-160	4.55	10.47	9.52	11.01	12.06	3.00	0.88	0.40
-192	4.75		11.76	12.01	13.38	3.10	0.90	0.40

SH 12293

NOTES:
1. Dimensions are in inches.
2. Material: MIL-B-24480
   O-ring - buna N

FIGURE 8. Male sil-braze tailpiece.

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Hose size (dash no.)	A (max)	B (max)	C +0.006 -0.000	D (nom)	E +0.006 -0.000
-80	7.31	1.62	6.63	6.13	0.72
-96	3.25	1.94	7.57	7.07	0.96
-128	10.78	2.38	9.97	9.47	0.80
-160	12.88	2.38	12.06	11.56	0.80
-192	14.98	2.38	13.38	12.88	0.80

SH 12294

NOTES:
1. Dimensions are in inches.
2. Material: Split clamp - QQ-C-390, alloy 955
   Band - QQ-N-281 and QQ-C-591, alloy 655
   Screws - QQ-C-591, alloy 655

FIGURE 9. Split clamp assembly.

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(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-H-24136/1(SH)	Hose, Synthetic Rubber, Synthetic Fiber Reinforced, And End Fittings, Reusable, For Flexible Hose Assemblies

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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